

React

Theory

1. Rendering Lists

- **Definition:** Use `.map()` to iterate over arrays and render JSX.
- **Key point:** Always provide a unique key prop for each element.

Example:

```
{arr.map((ele, index) => <h2 key={index}>{ele}</h2>)}
```

2. State Management

- **Definition:** `useState` hook manages component-level state.

Syntax:

```
const [data, setData] = useState(initialValue);
```

- Updating state triggers re-render.

3. Props

- **Definition:** Props are used to pass data from parent to child components.
- **Destructuring:**

```
const Child = ({result, count}) => <h1>{result} {count}</h1>;
```

- Props are **read-only**.

4. Component Communication

- **Parent → Child:** via props.
- **Child → Parent:** via callback functions passed as props.

Coding Round Questions

- Lists → How do you render lists in React? Why is the key prop important?
- State → Explain how `useState` works with an example.
- Props → What are props and how are they different from state?
- Parent-child communication → How do you pass data between parent and child components?
- Hooks → What are hooks and why were they introduced?

⚡ Machine Coding Round Challenge

Problem: Dynamic Student List

Requirements:

1. Create a parent component StudentList.
2. Maintain an array of student objects in state.
3. Render each student using a StudentCard child component.
4. Add a button to dynamically add a new student.
5. Pass props from parent to child for display.

```
import React, { useEffect, useState } from "react";

const List = () => {

  const arr = [1, 2, 3, 4, 5, 6, 7, 8,,10,11,22];

  const obj = [
    { name: "Arun", college: "Karan", Degree: "BSC" },
    { name: "Kumari", college: "Karan", Degree: "BCA" },
    { name: "Arun", college: "Karan", Degree: "BSC" },

    { name: "Arun", college: "Karan", Degree: "BSC" },

    { name: "Arun", college: "Karan", Degree: "BSC" },

    { name: "Arun", college: "Karan", Degree: "BSC" },
  ];

  return (
    <div>
      <h1>List</h1>

      {obj.map((ele, index) => {
        return (
          <div
            key={index}
            style={{
              border: "2px solid black",
              backgroundColor: "#333",
              color: "white",
              textAlign: "center",
              padding: "40px",
              margin: "10px",
            }}
          >
            {ele}
          </div>
        )
      })}
    </div>
  )
}
```

```

        }}
      >
        <h2>{ele.name}</h2>
        <h2>{ele.Degree}</h2>
        <h2>{ele.college}</h2>
      </div>
    );
  }}}

  {arr.map((ele, index) => (
    <h2 key={index}>{ele}</h2>
  ))}
</div>
);
};

export default List;

```

```

import { useState } from 'react'
import reactLogo from './assets/react.svg'
import viteLogo from './assets/vite.svg'
import heroImg from './assets/hero.png'
import './App.css'
import State from './components/State'
import List from './components/List'

function App() {

  return (
    <div>
      { /* <h2>Hello there</h2> */ }
      { /* <State/> */ }

      <List/>
    </div>
  )
}

export default App

```

```

import React from 'react'

const Child = ({result , count}) => {
  console.log(count);

```

```
    return (  
      <div>  
        <h1>{result}</h1>  
        <h1>{count}</h1>  
        <h2>Child component inside state </h2>  
      </div>  
    );  
  }  
  
export default Child
```